Large Language Models for Generative Recommendation and User Behavior Modeling

Abstract

The integration of Large Language Models (LLMs) into recommender systems has precipitated a paradigm shift from discriminative ranking to generative recommendation and sophisticated user behavior modeling. This survey comprehensively reviews the state-of-the-art in leveraging LLMs for generative recommendation, where items are predicted as token sequences, and for modeling complex, lifelong user behaviors through natural language interfaces. We synthesize findings from 300 relevant academic papers to construct a detailed taxonomy of methodological families, including ID tokenization strategies, retrieval-augmented generation (RAG) frameworks, instruction-tuned agents, and reinforcement learning-based alignment. We analyze how LLMs unify recall and ranking stages, mitigate data sparsity through world knowledge infusion, and enable explainable, controllable recommendations. Furthermore, we critically examine the challenges of inference latency, hallucination, bias amplification, and evaluation bottlenecks. By synthesizing empirical results and theoretical insights across diverse domains—from e-commerce and news to music and point-of-interest recommendation—this paper provides a rigorous foundation for understanding the current capabilities and future trajectories of LLM-driven recommendation systems.

1. Introduction

Recommender systems have traditionally relied on discriminative models that score and rank candidate items based on collaborative filtering signals or content features. While effective in dense data regimes, these approaches often struggle with cold-start problems, data sparsity, and the inability to leverage rich semantic context or world knowledge. The emergence of Large Language Models (LLMs) has introduced a transformative capability: the ability to model user preferences and item characteristics within a unified natural language space, enabling generative recommendation paradigms where items are predicted directly as text or token sequences [1], [2], [3].

The evolution of this field has moved rapidly from using LLMs as auxiliary feature extractors to deploying them as the core engine for end-to-end recommendation. Early approaches focused on enhancing traditional models with LLM-derived embeddings or using LLMs for post-hoc explanation generation [4], [5]. However, recent advancements have established "Generative Recommendation" as a distinct category, reformulating the recommendation task as a sequence generation problem akin to natural language processing [6], [7]. In this paradigm, user interaction histories are serialized into prompts, and the model autoregressively generates the next item identifier or description, effectively unifying retrieval and ranking into a single stage [8], [9].

Simultaneously, LLMs have revolutionized user behavior modeling. Traditional sequential recommenders often truncate long interaction histories due to context window limitations or rely on rigid ID-based representations that lack semantic interpretability. LLMs enable the modeling of lifelong user behaviors by compressing extensive interaction logs into scrutable natural language profiles, retrieving semantically relevant historical events, and simulating user dynamics through agent-based frameworks [10], [11], [12]. This capability allows for dynamic intent modeling, multi-interest disentanglement, and the integration of external world knowledge to infer latent preferences beyond observed interactions [13], [14].

Despite these advancements, significant challenges remain. The autoregressive nature of generative models introduces inference latency that conflicts with the real-time requirements of industrial systems [6], [15]. Hallucinations, where models generate non-existent items or factually incorrect explanations, pose risks to system reliability [3], [16]. Furthermore, biases inherent in pre-training corpora can propagate into recommendations, raising fairness concerns [17], [18]. This survey aims to provide a comprehensive overview of the methodological landscape, synthesizing evidence from 300 distinct studies to categorize approaches, compare performance, and identify open research directions. We organize the discussion around key methodological families, including tokenization strategies, retrieval augmentation, instruction tuning, and reinforcement learning, while critically assessing their limitations and deployment feasibility.

2. Method

The application of LLMs to recommendation and user behavior modeling encompasses a diverse array of architectures and training strategies. We categorize these methodologies into coherent families based on their core mechanisms: (1) Generative Recommendation via Item Tokenization, (2) Retrieval-Augmented and Memory-Enhanced Frameworks, (3) Instruction Tuning and Prompt Engineering, (4) Agent-Based and Tool-Learning Systems, (5) Reinforcement Learning and Preference Alignment, (6) User Behavior Simulation and Profiling, and (7) Efficiency and Deployment Optimizations.

2.1 Generative Recommendation via Item Tokenization

A fundamental challenge in generative recommendation is bridging the gap between the continuous, semantic space of LLMs and the discrete, collaborative signals of traditional recommender systems. This has led to the development of sophisticated item tokenization strategies that map items to sequences of tokens compatible with LLM vocabularies.

2.1.1 Semantic IDs and Vector Quantization One dominant approach involves replacing arbitrary numeric IDs with Semantic IDs (SIDs) derived from item content or collaborative structures. Methods like TIGER and LC-Rec employ Residual Quantized Variational Autoencoders (RQ-VAE) to discretize item embeddings into hierarchical token sequences [19], [20]. LC-Rec specifically utilizes tree-structured vector quantization on LLM-encoded text embeddings to generate conflict-free discrete indices, ensuring that semantically similar items share prefix tokens [20]. Similarly, TokenRec employs a Masked Vector-Quantized (MQ) Tokenizer to convert Graph Neural Network (GNN)-learned collaborative representations into discrete tokens, enabling the LLM to internalize high-order collaborative knowledge [21].

The efficacy of Semantic IDs lies in their ability to preserve neighborhood structures, facilitating better generalization to unseen items compared to atomic IDs [22]. However, scaling laws suggest that simply increasing model size in SID-based generative retrieval may not yield proportional improvements if the quantization bottleneck limits information capacity [23]. To address this, recent works like SETRec propose order-agnostic set identifiers that combine collaborative and semantic tokens, utilizing sparse attention masks to eliminate intra-item dependencies and enable parallel generation [24].

2.1.2 Direct Text Generation and ID Creation An alternative paradigm bypasses explicit ID tokenization by generating item titles or descriptions directly. GenRec reformulates recommendation as a sequence generation task using LLaMA, where the model predicts the next item title as text based on interaction history prompts [2]. This approach leverages the pre-trained semantic understanding of LLMs but faces challenges in output control and hallucination. To mitigate this, methods like SimGR bypass explicit token generation entirely, instead encoding user history and item semantics into a shared latent space and performing discriminative representation matching, thereby avoiding the biases inherent in autoregressive decoding [25].

Hybrid approaches attempt to combine the benefits of both worlds. TransRec constructs multi-facet identifiers that integrate numeric IDs, titles, and attributes, employing an FM-index for position-free constrained generation to ensure validity [26]. Similarly, GCRS factorizes the output into intent, item ID, and natural language response, using discrete semantic IDs to prevent vocabulary explosion while maintaining coherent dialogue [27]. Theoretical analyses confirm that under bijective mapping conditions, auto-regressive next-token prediction is equivalent to full-item-vocabulary maximum likelihood estimation, providing a formal basis for these generative paradigms [28].

2.1.3 Collaborative Signal Integration Purely semantic approaches often lack the precise collaborative signals captured by ID-based models. To address this, methods like E4SRec inject pre-trained SASRec item ID embeddings directly into the frozen LLM input space via linear projections, replacing the softmax vocabulary head with an item linear projection layer [29]. CLLM4Rec extends the LLM vocabulary with unique user and item ID tokens, employing mutually regularized pretraining to align collaborative and content-based predictions [30]. Furthermore, TCA4Rec projects item-level collaborative filtering logits into token-level distributions, allowing for soft-label alignment that balances behavioral fidelity with generative flexibility [31]. These hybrid strategies demonstrate that integrating collaborative signals is crucial for achieving state-of-the-art performance, particularly in dense interaction scenarios [32], [33].

2.2 Retrieval-Augmented and Memory-Enhanced Frameworks

Given the context window limitations of LLMs and the computational cost of processing long histories, Retrieval-Augmented Generation (RAG) and memory mechanisms have become essential for scaling LLM-based recommenders.

2.2.1 Semantic User Behavior Retrieval Traditional sequential models often truncate user histories to a fixed length, discarding valuable long-term preferences. ReLLa introduces Semantic User Behavior Retrieval (SUBR), which replaces chronological truncation with the retrieval of top-K semantically relevant items based on cosine similarity of LLM embeddings [10]. This approach significantly reduces noise and aligns the input context with the target item's relevance. ReLLaX further optimizes this by integrating SUBR with soft prompt augmentation, projecting conventional ID embeddings into the language space to homogenize input sequences [34].

Advanced retrieval strategies extend beyond simple similarity. AutoMR decouples memory storage from the LLM backbone, storing hidden representations of long-term interactions in an external memory bank and training a dedicated retriever using perplexity reduction as a proxy for relevance [35]. Similarly, GRASP employs generation-augmented retrieval, constructing offline semantic databases of LLM-generated item descriptions and applying multi-level holistic attention to fuse retrieved neighbors with user sequences [36]. These methods collectively demonstrate that semantic retrieval can effectively mitigate the "lost in the middle" phenomenon and enhance long-term interest modeling.

2.2.2 Knowledge-Augmented RAG Beyond user history, RAG frameworks integrate external world knowledge to enrich recommendations. KAR executes a three-stage pipeline involving factorization prompting to extract user preference reasoning and item factual knowledge, which are then adapted via hybrid-expert adapters [37]. ItemRAG shifts the retrieval focus from user-based to item-based co-purchase graphs, enabling robust retrieval even for cold-start items by leveraging semantic similarity expansions [38]. In conversational settings, CRAG employs a "Retrieve-Reflect-Rerank" pipeline where an LLM performs entity linking and reflection to filter collaborative retrieval results, significantly improving item coverage [39].

Domain-specific RAG applications further illustrate the versatility of this approach. In education, RAMO integrates course metadata into a vector store, allowing an LLM agent to retrieve and recommend MOOCs based on emotional intelligence prompts [40]. For food recommendation, an HEI-informed RAG framework retrieves candidate foods based on Healthy Eating Index scores, optimizing for nutritional improvements [41]. These systems highlight how RAG enables LLMs to access up-to-date, domain-specific knowledge without retraining, thereby mitigating hallucination and ensuring recommendation currency.

2.2.3 Memory Networks and Lifelong Modeling For modeling lifelong behaviors, memory networks provide a structured mechanism for storing and updating user profiles. MemoCRS integrates User-Specific Memory (UM) and General Memory (GM) into LLM prompts, employing an entity-based bank to store item attributes and user attitudes with timestamp tracking [42]. LIBER partitions lifelong sequences into short-term caches and long-term memory blocks, applying cascaded prompting to generate intra-block interest summaries and inter-block shift detections [43].

Hierarchical approaches like HiT-LBM partition behaviors into chunks and employ a Hierarchical Tree Search guided by process reward models to select optimal interest paths, effectively filtering hallucinated interests [44]. Similarly, PersonaX segments full historical behaviors via hierarchical clustering and dynamically allocates sampling budgets to preserve diversity, balancing information retention with input length constraints [45]. These memory-augmented architectures enable the modeling of complex, evolving user preferences over extended time horizons, a capability often absent in standard sequential models.

2.3 Instruction Tuning and Prompt Engineering

Instruction tuning has emerged as a powerful technique for aligning LLMs with recommendation tasks, enabling them to follow complex user intents and perform diverse recommendation scenarios.

2.3.1 Unified Instruction Formulations InstructRec formulates recommendation as an instruction-following task, employing a unified format that integrates preference, intention, task form, and context [46]. By generating 252K fine-grained personalized instructions using a teacher LLM, InstructRec bridges the gap between general language capabilities and specific recommendation goals. TALLRec adopts a two-stage tuning strategy, first performing Alpaca tuning for general instruction alignment and then conducting domain-specific "Rec-tuning" with minimal samples (<100), demonstrating that efficient adaptation is possible with limited data [47].

P5 (Pretrain, Personalized Prompt & Predict) takes this further by unifying diverse recommendation tasks (e.g., rating prediction, explanation generation, sequential recommendation) into a single text-to-text framework via personalized prompts [48]. This multitask pretraining allows for implicit knowledge transfer across tasks. Recent extensions like OpenP5 provide an open-source platform implementing these unified frameworks with various item indexing strategies, confirming that encoder-decoder architectures often outperform decoder-only models on sparse data due to parameter efficiency [49].

2.3.2 Prompt Engineering Strategies

Prompt engineering remains critical for zero-shot and few-shot performance. LLM-REC employs four prompting strategies, including engagement-guided summarization that leverages neighbor items to enrich incomplete descriptions [50]. For cold-start scenarios, LLMs can generate synthetic pairwise training signals by inferring latent preferences from textual descriptions, which are then integrated via auxiliary losses [51].

Advanced prompting techniques also address specific challenges. LANE utilizes a four-step Chain-of-Thought (CoT) prompting strategy to extract user multi-preferences and align them with black-box recommender features via attention mechanisms, enhancing performance without parameter updates [52]. In cross-domain settings, URLLM employs KNN retrieval to fetch similar user histories as few-shot prompts, enabling seamless knowledge transfer without embedding space alignment [53]. Furthermore, adaptive prompting frameworks like AdaptRec utilize LLM-driven active selection of collaborative signals, dynamically integrating similar user sequences as natural language demonstrations to bridge CF and LLM reasoning [54].

2.3.3 Preference Parsing and Profile Generation

Instruction tuning also facilitates explicit preference parsing. P2Rec reconstructs user prior preference distributions over latent item categories via user-level supervised fine-tuning, forcing the LLM to parse latent relationships without explicit labels [55]. LettinGo executes a three-stage feedback loop where diverse candidate profiles are generated via high-temperature sampling and evaluated based on downstream recommendation success, optimizing profile quality via Direct Preference Optimization (DPO) [56]. These methods illustrate how instruction tuning can transform LLMs from passive predictors into active reasoners capable of dissecting and articulating user preferences.

2.4 Agent-Based and Tool-Learning Systems

The agentic paradigm treats the LLM as an autonomous entity capable of planning, tool usage, and multi-turn interaction, moving beyond static prediction to dynamic decision-making.

2.4.1 Tool Learning and Function Calling

ToolRec deploys an LLM as a surrogate user that simulates decision-making via Chain-of-Thought prompting, iteratively selecting attributes and invoking external retrieval tools to refine recommendations [57]. TalkPlay-Tools orchestrates a sequential pipeline where an LLM agent predicts tool sequences (SQL, BM25, Dense, Generative) based on query type, enabling dynamic selection of appropriate retrieval modalities [58]. This modular approach allows the system to leverage specialized tools for specific sub-tasks, overcoming the limitations of monolithic LLMs.

In conversational settings, RecLLM employs a unified LLM for dialogue management, translating context tracking, preference extraction, and API calls into a single language modeling task without hardcoded policy graphs [59]. Similarly, Judy integrates hybrid routing logic to distinguish between structured SQL queries and unstructured RAG needs for outdoor trail recommendations, optimizing accuracy by retrieving relevant reviews rather than overwhelming the LLM [60]. These systems demonstrate that tool learning enables LLMs to interact with external databases and APIs, grounding their reasoning in factual data.

2.4.2 Multi-Agent and Reasoning Systems

Multi-agent frameworks further enhance reasoning capabilities. ReaSeq employs a hierarchical multi-agent system to distill user demand and product attributes into semantic embeddings, while a Diffusion LLM reconstructs plausible beyond-log behaviors [61]. GOT4Rec models recommendation as a directed graph of thoughts, decomposing user sequences into short-term, long-term, and collaborative preference nodes, and applying LLM-based voting aggregation to synthesize final recommendations [62].

Cognitive architectures like CogRec integrate Soar as a symbolic reasoning core with LLMs for semantic encoding, utilizing a bidirectional bridge module to translate natural language to production rules [63]. This neuro-symbolic integration allows for explainable, rule-based reasoning augmented by the flexibility of LLMs. Additionally, DeepRec orchestrates autonomous multi-turn interactions between LLMs and traditional recommendation models, using hierarchical RL rewards to guide deep item space exploration [64]. These advanced architectures highlight the potential of agentic systems to perform complex, multi-step reasoning tasks that exceed the capabilities of standard generative models.

2.5 Reinforcement Learning and Preference Alignment

Aligning LLM outputs with user preferences and business metrics often requires reinforcement learning (RL) or preference optimization techniques, moving beyond supervised fine-tuning.

2.5.1 Direct Preference Optimization (DPO)

Direct Preference Optimization (DPO) has gained traction for aligning LLMs with recommendation goals without explicit reward modeling. GQR employs iterative DPO with a BERT-based CTR predictor as a process reward model, aligning LLM generation with predicted click-through rates [65]. Similarly, LettinGo utilizes DPO to optimize user profile generation based on task-derived preferences, demonstrating that defining "good" profiles requires downstream feedback rather than static ground truth [56].

In conversational recommendation, DPO-Enhanced SumRec constructs preference pairs by selecting generated texts that minimize error against score predictors, jointly optimizing dialogue summaries and item descriptions [66]. For serendipity enhancement, SERAL trains SerenGPT using Collaborative Data Intervention (CDI) to construct high-quality preference pairs, optimized via Identity Preference Optimization (IPO) to burst filter bubbles [67]. These applications show that DPO provides a stable and efficient mechanism for aligning generative models with complex, multi-objective recommendation goals.

2.5.2 Reinforcement Learning from Feedback

Reinforcement Learning from Human Feedback (RLHF) and its variants are also widely used. GIRL employs a three-stage pipeline including Reward Model Training (RMT) on recruiter feedback and Proximal Policy Optimization (PPO) to align job recommendations with market demands [68]. REC-R1 formulates the LLM-RecSys interaction as a Markov Decision Process, directly optimizing LLM policies using black-box recommendation metrics (e.g., NDCG, Recall) as reward signals, eliminating the need for synthetic SFT data [69].

For query recommendation, AIGQ applies Interest-aware List GRPO (Group Relative Policy Optimization) with position-wise advantage estimation to optimize list diversity and intent density [70]. In proactive recommendation, LLM-IPP uses zero-shot prompting with constraint satisfaction to plan influence paths, leveraging world knowledge to construct semantically coherent bridges between user history and target items [71]. These RL-based approaches enable the optimization of long-term metrics and complex constraints that are difficult to capture via supervised learning alone.

2.5.3 Counterfactual and Causal Alignment Causal reasoning is increasingly integrated into RL frameworks to mitigate bias and improve robustness. CFT introduces a multi-task objective combining next-token prediction with a causal loss that minimizes cross-entropy between normal predictions and counterfactual predictions generated without history [72]. RoleGen employs counterfactual inference on intent anchors to model items as functional roles within a trajectory, capturing bilateral interaction information for dormant user re-engagement [73]. These methods illustrate how causal and counterfactual reasoning can enhance the robustness and fairness of RL-aligned recommenders.

2.6 User Behavior Simulation and Profiling

Accurate user behavior simulation and profiling are critical for evaluation, cold-start handling, and privacy-preserving recommendation.

2.6.1 LLM-Based User Simulators LLMs have proven effective as user simulators for offline evaluation and training. SimUSER executes a two-phase protocol involving self-consistent persona matching and interactive simulation with episodic memory, significantly reducing hallucination compared to baseline agents [74]. Lusifer operates via incremental profile updates with transparent explanation generation, leveraging unstructured textual metadata to simulate realistic feedback in data-scarce conditions [75].

For conversational evaluation, a five-task protocol decomposes simulation fidelity into dimensions like item distribution and preference correlation, revealing that simulators often skew towards popular items unless conditioned on specific constraints [76]. DyTA4Rec integrates dynamic profile updating and temporal pattern extraction via two-step prompting, enhancing simulation fidelity by explicitly modeling temporal dynamics [12]. These simulators enable cost-effective, large-scale evaluation and data augmentation, though care must be taken to mitigate bias amplification [74].

2.6.2 Natural Language User Profiles Natural language profiles offer interpretability and scrutability. Scrutable Natural Language Recommendation decomposes the task into profile generation and NL-based recommendation, allowing users to directly edit their preference summaries [11]. GenUP generates natural language user profiles from spatiotemporal check-in sequences using behavioral theories (e.g., Theory of Planned Behavior), enabling accurate next-POI prediction without extensive fine-tuning [77].

Automated profiling frameworks like APG4RecSim execute a three-stage pipeline for attribute extraction, context-aware consolidation, and counterfactual validation, distilling interaction history into coherent traits better than rigid manual schemas [78]. PURE employs a structured extraction pipeline to parse raw reviews into likes, dislikes, and features, iteratively refining profiles to manage token limits [79]. These approaches demonstrate that natural language profiles not only enhance interpretability but also serve as effective interfaces for user control and preference elicitation.

2.6.3 Multi-Interest and Latent Intent Modeling Modeling diverse user interests is crucial for comprehensive recommendation. LDMI operates at dual levels (user-individual and user-crowd), aligning LLM-derived semantic clusters with collaborative interest capsules to resolve granularity agnosticism [13]. Latent User Intent Models (LUIM) formulate intent as a latent variable mediating past context and future behavior, employing Conditional VAEs to infer intent from behavior signals without manual definition [80].

SPECTRA reframes user modeling as distributional inference over interpretable semantic clusters, extracting explicit posterior distributions via logit probing to mitigate exposure bias and head concentration [81]. These methods enable the disentanglement of multifaceted user interests, supporting diverse and serendipitous recommendations that cater to different aspects of a user's persona.

2.7 Efficiency and Deployment Optimizations

Deploying LLMs in production environments requires addressing challenges related to latency, memory, and computational cost.

2.7.1 Speculative Decoding and Acceleration Speculative decoding techniques have been adapted for recommendation to accelerate inference. AtSpeed addresses the N-to-K verification challenge by training a draft model to align top-K sequences with the target LLM, achieving significant speedups while maintaining accuracy [82]. LASER partitions users/items into groups to construct compact, personalized Trie-based retrieval pools, employing relaxed verification to accept top-k probable tokens and achieving up to 4.77x speedup [83].

Token compression methods like TCRA-LLM integrate dense/sparse retrieval with post-retrieval compression, utilizing fine-tuned summarization to preserve critical semantics while reducing token counts by 65% [84]. STAMP employs Semantic Adaptive Pruning to dynamically prune redundant tokens mid-forward pass based on semantic saliency, reducing training time and VRAM usage [85]. These optimizations are essential for making LLM-based recommenders viable in real-time industrial settings.

2.7.2 Parameter-Efficient Fine-Tuning (PEFT) PEFT techniques like LoRA and adapters are widely used to adapt large models efficiently. Instance-wise LoRA (iLoRA) decomposes standard LoRA matrices into expert sub-matrices, utilizing a gating network to generate instance-specific weights for granular behavior modeling [86]. RecLoRA integrates a Personalized LoRA module that generates user-specific low-rank matrices via soft routing guided by a pre-trained conventional recommendation model, enabling lifelong personalization with minimal parameters [87].

Federated adaptation frameworks like FedPA initialize clients with a pre-trained foundation model and learn lightweight low-rank adapters at user and group levels, balancing personalization with privacy [88]. CE-LoRA further reduces communication costs in federated settings by transmitting only a compact full-rank intermediate matrix, achieving 1024x communication reduction [89]. These methods demonstrate that efficient adaptation is key to scaling LLM recommenders across diverse users and devices.

2.7.3 Cloud-Edge Collaboration Hybrid cloud-edge architectures address latency and data staleness. LSC4Rec deploys LLMs on the cloud for candidate generation using historical data and Small Recommendation Models (SRMs) on devices for real-time reranking, employing collaborative training to align SRM distributions with LLM outputs [90]. This division of labor leverages the generalization power of cloud LLMs while maintaining the responsiveness of edge models, offering a practical solution for device-cloud recommendation scenarios.

3. Challenges and Outlook

Despite significant progress, the field of LLM-based generative recommendation and user behavior modeling faces several critical challenges that must be addressed to realize its full potential.

3.1 Inference Latency and Scalability

The autoregressive nature of generative models inherently introduces latency, which conflicts with the sub-100ms Service Level Agreements (SLAs) required by many industrial applications [6], [15]. While speculative decoding and quantization offer partial solutions, scaling to millions of items remains challenging due to memorization constraints and the computational cost of beam search [8], [82]. Future research must focus on developing non-autoregressive generative paradigms, more efficient tokenization schemes, and hybrid architectures that balance generative flexibility with discriminative efficiency.

3.2 Hallucination and Reliability

Hallucination, where models generate non-existent items or factually incorrect explanations, poses a significant risk to user trust and system reliability [3], [16]. While RAG and constrained decoding mitigate this issue, they do not eliminate it entirely, especially in open-world scenarios [37]. Robust verification mechanisms, fact-checking modules, and uncertainty estimation techniques are needed to ensure the factual accuracy of generated recommendations and explanations.

3.3 Bias and Fairness

LLMs inherit biases from their pre-training corpora, which can propagate into recommendations, leading to demographic, cultural, and popularity biases [17], [18]. Studies have shown that LLMs may reinforce stereotypes or exhibit disparate impact across different user groups [91]. Mitigation strategies such as debiasing prompts, fair retrieval mechanisms, and adversarial training are essential, but a comprehensive framework for evaluating and ensuring fairness in generative recommenders is still lacking.

3.4 Evaluation Bottlenecks

Traditional metrics like NDCG and Recall may not fully capture the quality of generative recommendations, particularly regarding diversity, serendipity, and explainability [16], [92]. New evaluation protocols that assess semantic coherence, user satisfaction, and long-term engagement are needed. Furthermore, the reliance on offline datasets may not reflect real-world dynamics, necessitating more robust online A/B testing frameworks and user studies [76], [93].

3.5 Privacy and Security

The use of LLMs raises privacy concerns, particularly when processing sensitive user data or deploying models in federated settings [94]. Additionally, LLM-based systems are vulnerable to adversarial attacks, such as data poisoning via textual rewriting, which can subtly shift item rankings [95]. Developing privacy-preserving training methods, secure inference protocols, and robust defense mechanisms against adversarial attacks is crucial for safe deployment.

3.6 Future Directions

Future research should explore the integration of multimodal data (images, audio, video) into generative frameworks, leveraging the increasing capabilities of multimodal LLMs [96], [97]. The development of foundation models specifically tailored for recommendation, trained on massive behavioral corpora, could further bridge the gap between general language understanding and domain-specific expertise [9], [98]. Finally, the emergence of autonomous agents capable of proactive interaction and long-term planning promises to transform recommender systems from passive filters into active assistants [99], [100].

4. Conclusion

The integration of Large Language Models into recommender systems has ushered in a new era of generative recommendation and advanced user behavior modeling. By reformulating recommendation as a sequence generation task, leveraging retrieval-augmented architectures, and employing instruction tuning and reinforcement learning, researchers have developed systems that are more interpretable, flexible, and capable of leveraging world knowledge than their discriminative predecessors. This survey has synthesized evidence from 300 academic papers to provide a comprehensive overview of the methodological landscape, highlighting key innovations in tokenization, memory mechanisms, agent-based reasoning, and efficiency optimization.

While challenges related to latency, hallucination, bias, and evaluation remain, the rapid pace of innovation suggests that these hurdles will be progressively overcome. The future of recommendation lies in the seamless integration of LLMs with traditional collaborative filtering signals, the development of robust and fair generative frameworks, and the emergence of autonomous agents that can actively engage with users to satisfy their evolving needs. As the field matures, LLM-driven recommenders are poised to become the standard for personalized information access, transforming how users discover and interact with content in the digital age.

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